

Remove the log factor in residual complexity for locally Lipschitz monotone inclusions

ChatGPT (gpt-5.6-sol)*

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Abstract

We consider the monotone inclusion

$$0 \in (F + B)(x)$$

in a finite-dimensional Euclidean space, where B is maximal monotone, $F + B$ is monotone and has a zero, and F is locally Lipschitz on $\text{cl}(\text{dom } B)$. We present an anchored splitting method with retained backtracking. Each trial uses two evaluations of F and two exact resolvent evaluations of B , and every accepted iterate y_k is accompanied by an explicitly computable graph element $d_k \in (F + B)(y_k)$.

A weighted Lyapunov identity yields the last-iterate estimate

$$\|d_k\| \leq \frac{H}{\Lambda_k}, \quad \Lambda_k = \sum_{i=1}^k \lambda_i,$$

for an instance-dependent constant $H < \infty$. A uniform analysis of all accepted and rejected trial points shows that the primary stepsizes satisfy $\lambda_k \geq \lambda_* > 0$; consequently,

$$\text{res}_{F+B}(y_k) \leq \frac{H}{k\lambda_*}.$$

The primary line search performs only a constant number of reductions, while the secondary line search performs $O(\log N)$ reductions over the first N accepted iterations. The total number of trials is therefore $N + O(\log N)$, and an ε -residual solution is found using $O(\varepsilon^{-1})$ evaluations of F and exact resolvent evaluations of B .

This answers the question of Lu and Mei concerning the removal of the logarithmic factor in the non-strongly monotone case, under their oracle convention that $J_{\gamma B}$ has unit cost for every $\gamma > 0$. The result does not address inexact resolvent evaluations or numerical bit complexity.

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1 The open problem

This paper addresses an open problem posed by Zhaosong Lu, Sanyou Mei in *Primal-Dual Extrapolation Methods for Monotone Inclusions Under Local Lipschitz Continuity* [1] (Mathematics of Operations Research, 2024), Section 7 (Concluding remarks), page 23. The website catalogue entry is problem 134 (W4403279582_p0) of the [Open Problems in Operations Research](#) collection.

Design an algorithm for the above monotone inclusion setting (monotone F , maximal monotone B , and F locally Lipschitz continuous on $\overline{\text{dom } B}$) that finds an ε -residual solution $x \in \text{dom } B$ with $\text{res}_{F+B}(x) \leq \varepsilon$ using at most $O(\varepsilon^{-1})$ evaluations of F and $(I + \gamma B)^{-1}$, i.e., improve the known $O(\varepsilon^{-1} \log \varepsilon^{-1})$ operation complexity to $O(\varepsilon^{-1})$ in this local-Lipschitz setting.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

Let $B : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ be maximal monotone and let $F : \text{cl}(\text{dom } B) \rightarrow \mathbb{R}^n$ be locally Lipschitz. We study the problem

$$\text{find } x \in \mathbb{R}^n \quad \text{such that} \quad 0 \in (F + B)(x). \quad (1)$$

For a point $x \in \text{dom } B$, the residual is defined by

$$\text{res}_{F+B}(x) := \inf \{ \|v\| : v \in (F + B)(x) \}.$$

Lu and Mei [3] established operation bounds of $O(\log \varepsilon^{-1})$ and $O(\varepsilon^{-1} \log \varepsilon^{-1})$ for the strongly and non-strongly monotone cases, respectively. Their operation model counts evaluations of F and exact evaluations of the resolvent

$$J_{\gamma B} := (I + \gamma B)^{-1}$$

for arbitrary $\gamma > 0$. In their concluding remarks, they asked whether the latter bound can be improved to $O(\varepsilon^{-1})$, matching the bound known under global Lipschitz continuity.

The method presented below achieves this bound. Its analysis rests on three main ingredients. First, an anchored update produces a last-iterate Lyapunov identity in which monotonicity of $F + B$ supplies the correct sign. Second, a summable tolerance sequence absorbs the error introduced by an auxiliary resolvent. Third, retained backtracking, together with a uniform bound on all trial points, implies a positive lower bound on the primary stepsize and only logarithmically many reductions of the auxiliary resolvent parameter.

Importantly, the computable vector d_k used in the stopping test is a graph element of $F + B$, not necessarily the minimum-norm element of $(F + B)(y_k)$. It therefore provides the rigorous upper bound

$$\text{res}_{F+B}(y_k) \leq \|d_k\|.$$

3 Problem setting and algorithm

We impose the following assumptions.

Assumption 1. The following conditions hold.

1. $B : \mathbb{R}^n \rightrightarrows \mathbb{R}^n$ is maximal monotone.
2. $\mathcal{T} := F + B$ is monotone on $\text{dom } B$.
3. The solution set $\text{zer } \mathcal{T} := \{p \in \mathbb{R}^n : 0 \in \mathcal{T}(p)\}$ is nonempty.
4. $F : \text{cl}(\text{dom } B) \rightarrow \mathbb{R}^n$ is locally Lipschitz. Equivalently, in the finite-dimensional setting considered here: for every compact set $K \subseteq \text{cl}(\text{dom } B)$ there exists $L_K < \infty$ such that

$$\|F(x) - F(y)\| \leq L_K \|x - y\|, \quad x, y \in K.$$

5. $J_{\gamma B}$ can be evaluated exactly for every $\gamma > 0$, with each evaluation assigned unit oracle cost.

No separate monotonicity assumption on F is needed in the proofs; only monotonicity of the sum $\mathcal{T} = F + B$ is used.

Fix an anchor $u \in \mathbb{R}^n$, upper parameters $\bar{\lambda}, \bar{\eta} > 0$, contraction factors $\rho_\lambda, \rho_\eta \in (0, 1)$, and a constant $\Delta > 0$. Set

$$\delta_k := \frac{\Delta}{(k+1)^2}, \quad k \geq 1. \quad (2)$$

Algorithm 1 (Retained-backtracking anchored splitting). Given a target accuracy $\varepsilon > 0$, initialize

$$\Lambda_0 = 0, \quad y_0 = u, \quad d_0 = 0, \quad \lambda = \bar{\lambda}, \quad \eta = \bar{\eta}.$$

For $k = 1, 2, \dots$, repeat the following trial until it is accepted.

1. Form the provisional quantities

$$\Lambda = \Lambda_{k-1} + \lambda, \quad \tau = \frac{\lambda}{\Lambda}, \quad c = \frac{\Lambda_{k-1}}{\Lambda} = 1 - \tau. \quad (3)$$

2. Compute

$$a = \tau u + c y_{k-1}, \quad z = a - \lambda c d_{k-1}, \quad (4)$$

$$x = J_{\eta B}(z), \quad y = J_{\lambda B}(a - \lambda F(x)). \quad (5)$$

Evaluate $F(x)$ and $F(y)$, and define

$$e := F(x) - F(y), \quad (6)$$

$$b := \frac{a - y}{\lambda} - F(x), \quad d := b + F(y), \quad (7)$$

$$r := \frac{z - y}{\lambda}. \quad (8)$$

3. Check the two tests

$$\lambda \|F(x) - F(y)\| \leq \frac{1}{4} \|x - y\|, \quad (\text{L})$$

$$\|x - y\| \leq 2\|z - y\| + 4\lambda\delta_k. \quad (\text{P})$$

4. If (L) fails, set $\lambda \leftarrow \rho_\lambda \lambda$ and restart the trial. If (L) holds but (P) fails, set $\eta \leftarrow \rho_\eta \eta$ and restart the trial.

Rejected trials do not modify $(\Lambda_{k-1}, y_{k-1}, d_{k-1})$. In particular, after a reduction of λ , all provisional quantities in (3)–(8) are recomputed.

5. If both tests hold, accept the trial and set

$$(\Lambda_k, \tau_k, c_k, \lambda_k, \eta_k, a_k, z_k, x_k, y_k, e_k, b_k, d_k, r_k) := (\Lambda, \tau, c, \lambda, \eta, a, z, x, y, e, b, d, r).$$

The accepted values of λ and η are retained as the starting values for the next iteration; they are never reset or increased.

Terminate and return y_k if

$$\|d_k\| \leq \varepsilon. \quad (9)$$

Every full trial uses exactly two resolvent evaluations,

$$J_{\eta B}(z) \quad \text{and} \quad J_{\lambda B}(a - \lambda F(x)),$$

and two evaluations of F , namely $F(x)$ and $F(y)$. All quantities appearing in the two tests are then available without further oracle calls.

4 Graph certificate and exact identities

We begin with identities that hold at every accepted iteration.

Lemma 1 (Graph certificate). *For every accepted iteration k ,*

$$b_k \in B(y_k), \quad d_k \in \mathcal{T}(y_k).$$

Consequently,

$$\text{res}_{\mathcal{T}}(y_k) \leq \|d_k\|.$$

Proof. The definition

$$y_k = J_{\lambda_k B}(a_k - \lambda_k F(x_k))$$

is equivalent to

$$a_k - \lambda_k F(x_k) \in y_k + \lambda_k B(y_k).$$

Therefore,

$$b_k = \frac{a_k - y_k}{\lambda_k} - F(x_k) \in B(y_k),$$

and it follows that

$$d_k = b_k + F(y_k) \in (F + B)(y_k) = \mathcal{T}(y_k).$$

The residual inequality is immediate from the definition of the residual. $\square$

The certificate d_k need not be the minimum-norm element of $\mathcal{T}(y_k)$. This distinction is immaterial for the stopping rule, because a graph element always provides an upper bound on the residual.

Lemma 2 (Basic identities). *For every accepted $k \geq 1$,*

$$\frac{a_k - y_k}{\lambda_k} = d_k + e_k, \quad (10)$$

$$r_k = d_k - c_k d_{k-1} + e_k, \quad (11)$$

$$y_k - y_{k-1} = \tau_k(u - y_{k-1}) - \lambda_k(d_k + e_k). \quad (12)$$

Thus, with $q_k := d_k - c_k d_{k-1}$,

$$r_k = q_k + e_k.$$

Proof. By the definitions of b_k , d_k , and e_k ,

$$\begin{aligned} d_k &= \frac{a_k - y_k}{\lambda_k} - F(x_k) + F(y_k) \\ &= \frac{a_k - y_k}{\lambda_k} - e_k, \end{aligned}$$

which proves (10). Since $z_k = a_k - \lambda_k c_k d_{k-1}$,

$$\begin{aligned} r_k &= \frac{z_k - y_k}{\lambda_k} \\ &= \frac{a_k - y_k}{\lambda_k} - c_k d_{k-1} \\ &= d_k - c_k d_{k-1} + e_k, \end{aligned}$$

which proves (11). Finally, $a_k = \tau_k u + c_k y_{k-1}$ and $c_k = 1 - \tau_k$, so (10) gives

$$\begin{aligned} y_k - y_{k-1} &= a_k - y_{k-1} - \lambda_k(d_k + e_k) \\ &= \tau_k(u - y_{k-1}) - \lambda_k(d_k + e_k). \end{aligned} \quad \square$$

The two acceptance tests provide precisely the error estimate needed below.

Lemma 3 (Accepted-error estimate). *At every accepted iteration,*

$$\|e_k\| \leq \frac{1}{2}\|r_k\| + \delta_k. \quad (13)$$

Consequently,

$$\|r_k\|^2 - \|e_k\|^2 \geq \frac{1}{2}\|r_k\|^2 - 2\delta_k^2 \geq -2\delta_k^2. \quad (14)$$

Proof. Tests (L) and (P) imply

$$\begin{aligned} \lambda_k \|e_k\| &\leq \frac{1}{4}\|x_k - y_k\| \\ &\leq \frac{1}{2}\|z_k - y_k\| + \lambda_k \delta_k \\ &= \lambda_k \left(\frac{1}{2}\|r_k\| + \delta_k \right). \end{aligned}$$

Dividing by $\lambda_k > 0$ proves (13). Moreover, for all $s, \delta \geq 0$,

$$\left(\frac{1}{2}s + \delta\right)^2 \leq \frac{1}{2}s^2 + 2\delta^2,$$

because the difference between the right- and left-hand sides equals $(s/2 - \delta)^2$. Applying this inequality with $s = \|r_k\|$ proves (14). $\square$

5 Lyapunov identity and the residual rate

Recall that

$$\Lambda_k = \sum_{i=1}^k \lambda_i, \quad \tau_k = \frac{\lambda_k}{\Lambda_k}, \quad c_k = \frac{\Lambda_{k-1}}{\Lambda_k}.$$

Define

$$\ell_k := \langle d_k, u - y_k \rangle - \frac{\Lambda_k}{2} \|d_k\|^2, \quad k \geq 1, \quad (15)$$

and, for convenience, set $\ell_0 = 0$.

Lemma 4 (Exact one-step identity). *For every $k \geq 2$,*

$$\begin{aligned} \ell_k - c_k \ell_{k-1} &= \frac{\Lambda_k}{2} (\|r_k\|^2 - \|e_k\|^2) \\ &\quad + \frac{c_k}{\tau_k} \langle y_k - y_{k-1}, d_k - d_{k-1} \rangle, \end{aligned} \quad (16)$$

and the last inner product is nonnegative. At $k = 1$,

$$\ell_1 = \frac{\Lambda_1}{2} (\|r_1\|^2 - \|e_1\|^2). \quad (17)$$

Consequently, for every $k \geq 1$,

$$\ell_k - c_k \ell_{k-1} \geq \frac{\Lambda_k}{2} (\|r_k\|^2 - \|e_k\|^2). \quad (18)$$

Proof. Let

$$h_k := y_k - y_{k-1}.$$

By (12),

$$u - y_{k-1} = \frac{h_k + \lambda_k(d_k + e_k)}{\tau_k}.$$

Therefore,

$$\begin{aligned} u - y_k &= u - y_{k-1} - h_k \\ &= \frac{c_k}{\tau_k} h_k + \Lambda_k(d_k + e_k), \end{aligned}$$

where we used $\lambda_k/\tau_k = \Lambda_k$. It follows that

$$\ell_k = \frac{c_k}{\tau_k} \langle h_k, d_k \rangle + \frac{\Lambda_k}{2} \|d_k\|^2 + \Lambda_k \langle e_k, d_k \rangle. \quad (19)$$

Similarly, since $\Lambda_{k-1} = c_k \Lambda_k$,

$$\begin{aligned} c_k \ell_{k-1} &= \frac{c_k}{\tau_k} \langle h_k, d_{k-1} \rangle + c_k \Lambda_k \langle d_{k-1}, d_k + e_k \rangle \\ &\quad - \frac{c_k^2 \Lambda_k}{2} \|d_{k-1}\|^2. \end{aligned} \tag{20}$$

Subtracting (20) from (19) gives

$$\begin{aligned} \ell_k - c_k \ell_{k-1} &= \frac{c_k}{\tau_k} \langle h_k, d_k - d_{k-1} \rangle \\ &\quad + \frac{\Lambda_k}{2} \|d_k - c_k d_{k-1}\|^2 \\ &\quad + \Lambda_k \langle d_k - c_k d_{k-1}, e_k \rangle. \end{aligned}$$

Using

$$r_k = d_k - c_k d_{k-1} + e_k$$

together with the identity

$$\|v + e\|^2 - \|e\|^2 = \|v\|^2 + 2 \langle v, e \rangle,$$

we obtain (16).

For $k \geq 2$, Lemma 1 gives

$$(y_k, d_k), (y_{k-1}, d_{k-1}) \in \text{gph } \mathcal{T}.$$

Monotonicity of $\mathcal{T}$ therefore implies

$$\langle y_k - y_{k-1}, d_k - d_{k-1} \rangle \geq 0.$$

At $k = 1$, the formal pair $(y_0, d_0) = (u, 0)$ need not belong to $\text{gph } \mathcal{T}$, so monotonicity cannot be invoked. Instead, $c_1 = 0$, $\tau_1 = 1$, and

$$u - y_1 = \lambda_1(d_1 + e_1), \quad r_1 = d_1 + e_1.$$

Thus

$$\begin{aligned} \ell_1 &= \lambda_1 \langle d_1, d_1 + e_1 \rangle - \frac{\lambda_1}{2} \|d_1\|^2 \\ &= \frac{\lambda_1}{2} (\|d_1 + e_1\|^2 - \|e_1\|^2), \end{aligned}$$

which is (17). □

Combining the preceding two lemmas yields the fundamental recurrence.

Corollary 1 (Weighted Lyapunov recurrence). *For every accepted $k \geq 1$,*

$$\ell_k - c_k \ell_{k-1} \geq \frac{\Lambda_k}{4} \|r_k\|^2 - \Lambda_k \delta_k^2, \tag{21}$$

$$\Lambda_k \ell_k \geq \Lambda_{k-1} \ell_{k-1} - \Lambda_k^2 \delta_k^2. \tag{22}$$

Proof. Inequality (21) follows from (18) and (14). Dropping the nonnegative term involving r_k and multiplying by Λ_k gives (22), since

$$\Lambda_k c_k = \Lambda_{k-1}. \quad \square$$

Fix any solution $p \in \text{zer } \mathcal{T}$ and introduce the constants

$$D := \|u - p\|, \quad (23)$$

$$S_0 := \bar{\lambda}^2 \Delta^2 \sum_{i=1}^{\infty} \frac{i^2}{(i+1)^4} < \infty, \quad (24)$$

$$H := D + \sqrt{D^2 + 2S_0}. \quad (25)$$

The series in (24) may also be written as

$$\sum_{i=1}^{\infty} \frac{i^2}{(i+1)^4} = \zeta(2) - 2\zeta(3) + \zeta(4).$$

Proposition 1 (Last-iterate graph residual). *Every accepted iterate satisfies*

$$\|d_k\| \leq \frac{H}{\Lambda_k}. \quad (26)$$

Hence

$$\text{res}_{\mathcal{T}}(y_k) \leq \frac{H}{\Lambda_k}. \quad (27)$$

Proof. Since every accepted stepsize satisfies $\lambda_i \leq \bar{\lambda}$, we have

$$\Lambda_i \leq i\bar{\lambda}.$$

Consequently,

$$\sum_{i=1}^{\infty} \Lambda_i^2 \delta_i^2 \leq \bar{\lambda}^2 \Delta^2 \sum_{i=1}^{\infty} \frac{i^2}{(i+1)^4} = S_0.$$

Telescoping (22), with $\Lambda_0 \ell_0 = 0$, gives

$$\Lambda_k \ell_k \geq -S_0. \quad (28)$$

Since $d_k \in \mathcal{T}(y_k)$ and $0 \in \mathcal{T}(p)$, monotonicity gives

$$\langle d_k, y_k - p \rangle \geq 0.$$

It follows that

$$\begin{aligned} \ell_k &= \langle d_k, u - y_k \rangle - \frac{\Lambda_k}{2} \|d_k\|^2 \\ &= \langle d_k, u - p \rangle - \langle d_k, y_k - p \rangle - \frac{\Lambda_k}{2} \|d_k\|^2 \\ &\leq D \|d_k\| - \frac{\Lambda_k}{2} \|d_k\|^2. \end{aligned}$$

Combining this inequality with (28) and setting $t_k := \Lambda_k \|d_k\|$ gives

$$\frac{1}{2} t_k^2 - D t_k - S_0 \leq 0.$$

The nonnegative root of the associated quadratic is

$$D + \sqrt{D^2 + 2S_0} = H.$$

Thus $t_k \leq H$, which proves (26). Estimate (27) then follows from Lemma 1. $\square$

6 Uniform backtracking analysis

The preceding conclusions hold for every finite accepted prefix, without presuming that future line searches terminate. We now use these prefix-uniform estimates to establish well-definedness and uniform complexity bounds without circularity.

First, (26) implies

$$\lambda_k \|d_k\| \leq H\tau_k, \quad (29)$$

$$\lambda_k c_k \|d_{k-1}\| \leq H\tau_k \quad (k \geq 2). \quad (30)$$

Indeed, $\lambda_k/\Lambda_k = \tau_k$ gives the first inequality, while

$$\begin{aligned} \lambda_k c_k \|d_{k-1}\| &\leq \lambda_k \frac{\Lambda_{k-1}}{\Lambda_k} \frac{H}{\Lambda_{k-1}} \\ &= H\tau_k \end{aligned}$$

gives the second.

The accepted-error estimate and $r_k = d_k - c_k d_{k-1} + e_k$ also yield

$$\|e_k\| \leq \frac{1}{2} (\|d_k\| + c_k \|d_{k-1}\| + \|e_k\|) + \delta_k.$$

Therefore,

$$\lambda_k \|e_k\| \leq 2H\tau_k + 2\lambda_k \delta_k. \quad (31)$$

Lemma 5 (Uniform bounds for accepted and trial points). *There is a bounded set, independent of the iteration number and of the number of rejected trials, that contains every accepted iterate y_k and every trial point x and y generated by Algorithm 1.*

Proof. Using (12), (29), and (31),

$$\begin{aligned} \|y_k - p\| &\leq c_k \|y_{k-1} - p\| + \tau_k D + \lambda_k \|d_k\| + \lambda_k \|e_k\| \\ &\leq c_k \|y_{k-1} - p\| + \tau_k (D + 3H) + 2\lambda_k \delta_k. \end{aligned}$$

Moreover,

$$\begin{aligned} 2\lambda_k \delta_k &= 2\tau_k \Lambda_k \frac{\Delta}{(k+1)^2} \\ &\leq 2\tau_k \frac{k\bar{\lambda}\Delta}{(k+1)^2} \\ &\leq 2\tau_k \bar{\lambda}\Delta. \end{aligned}$$

Set

$$C_y := D + 3H + 2\bar{\lambda}\Delta. \quad (32)$$

Since $y_0 = u$ and $\|u - p\| = D \leq C_y$, induction gives

$$\|y_k - p\| \leq C_y \quad \text{for every accepted } k. \quad (33)$$

Now consider an arbitrary trial at iteration k after a finite accepted prefix; we suppress trial superscripts. Since $a = \tau u + c y_{k-1}$,

$$\|a - p\| \leq C_y.$$

For $k \geq 2$,

$$\begin{aligned}\lambda c \|d_{k-1}\| &\leq \lambda \frac{\Lambda_{k-1}}{\Lambda_{k-1} + \lambda} \frac{H}{\Lambda_{k-1}} \\ &= H \frac{\lambda}{\Lambda_{k-1} + \lambda} = H\tau \leq H.\end{aligned}$$

The same bound holds trivially at $k = 1$, since $c = 0$. Hence

$$\|z - p\| \leq C_y + H. \quad (34)$$

Since $p \in \text{zer } \mathcal{T}$, we have $-F(p) \in B(p)$, and therefore, for every $\eta > 0$,

$$p = J_{\eta B}(p - \eta F(p)).$$

Nonexpansiveness of the resolvent gives

$$\begin{aligned}\|x - p\| &\leq \|z - p + \eta F(p)\| \\ &\leq C_y + H + \bar{\eta} \|F(p)\| =: R_x.\end{aligned} \quad (35)$$

Since $x \in \text{dom } B$, every trial point x lies in the compact set

$$K_x := \text{cl}(\text{dom } B) \cap \{v : \|v - p\| \leq R_x\}.$$

By local Lipschitz continuity, F is continuous on K_x , so

$$G_x := \max_{v \in K_x} \|F(v)\| < \infty.$$

Likewise,

$$p = J_{\lambda B}(p - \lambda F(p)),$$

and hence

$$\begin{aligned}\|y - p\| &\leq \|a - p - \lambda(F(x) - F(p))\| \\ &\leq C_y + \bar{\lambda}(G_x + \|F(p)\|) =: R_y.\end{aligned} \quad (36)$$

Thus all accepted iterates and all trial points x and y lie in a fixed bounded subset of $\text{dom } B$. $\square$

Let

$$R := \max\{C_y, R_x, R_y\}, \quad K := \text{cl}(\text{dom } B) \cap \{v : \|v - p\| \leq R\}.$$

The set K is compact. By local Lipschitz continuity, there is a finite constant $L_* \geq 0$ such that

$$\|F(v) - F(w)\| \leq L_* \|v - w\|, \quad v, w \in K. \quad (37)$$

Lemma 6 (Primary line search). *The total number of reductions of λ is bounded by the instance-dependent constant*

$$m_\lambda := \left\lceil \frac{\log(\max\{1, 4\bar{\lambda}L_*\})}{\log(1/\rho_\lambda)} \right\rceil. \quad (38)$$

In particular, every trial and accepted primary stepsize satisfies

$$\lambda \geq \lambda_* := \bar{\lambda} \rho_\lambda^{m_\lambda} > 0. \quad (39)$$

Proof. For any trial pair x, y , (37) gives

$$\lambda \|F(x) - F(y)\| \leq \lambda L_* \|x - y\|.$$

Thus test (L) necessarily holds whenever

$$\lambda \leq \frac{1}{4L_*},$$

with the convention that this condition is automatic if $L_* = 0$.

The candidate λ is initialized at $\bar{\lambda}$, is multiplied by ρ_λ after each failure of (L), and is never increased. By the definition of m_λ ,

$$4L_* \bar{\lambda} \rho_\lambda^{m_\lambda} \leq 1.$$

Hence test (L) cannot fail after more than m_λ reductions. This also proves (39). $\square$

The positive lower bound on λ now supplies a uniform bound on the graph element of B produced by every trial point y . Indeed,

$$b_y := \frac{a - y}{\lambda} - F(x) \in B(y)$$

and

$$\|b_y\| \leq \frac{C_y + R_y}{\lambda_*} + G_x =: M < \infty. \quad (40)$$

Lemma 7 (Secondary line search). *The secondary line search terminates at every iteration. Over the first N accepted iterations, the total number of reductions of η is at most*

$$m_\eta(N) := \left\lceil \frac{\log \left(\max \left\{ 1, \frac{\bar{\eta} M (N+1)^2}{4\lambda_* \Delta} \right\} \right)}{\log(1/\rho_\eta)} \right\rceil = O(\log(N+1)). \quad (41)$$

Proof. For a trial at iteration k , we have

$$x = J_{\eta B}(z).$$

Since $b_y \in B(y)$,

$$y = J_{\eta B}(y + \eta b_y).$$

Nonexpansiveness of $J_{\eta B}$ and (40) give

$$\begin{aligned} \|x - y\| &\leq \|z - y - \eta b_y\| \\ &\leq \|z - y\| + \eta M. \end{aligned} \quad (42)$$

Therefore, test (P) necessarily holds whenever

$$\eta M \leq 4\lambda \delta_k. \quad (43)$$

Indeed, under (43),

$$\|x - y\| \leq \|z - y\| + 4\lambda \delta_k \leq 2\|z - y\| + 4\lambda \delta_k.$$

For $k \leq N$,

$$4\lambda \delta_k \geq \frac{4\lambda_* \Delta}{(N+1)^2}.$$

After $m_\eta(N)$ cumulative reductions, the retained parameter satisfies

$$\eta M \leq \frac{4\lambda_*\Delta}{(N+1)^2},$$

and at that point (43) holds at every iteration $k \leq N$. Because η is retained rather than reset, no more than $m_\eta(N)$ reductions can occur through iteration N . $\square$

7 Main complexity result

Theorem 1 ($O(\varepsilon^{-1})$ operation complexity). *Under Assumption 1, Algorithm 1 is well-defined. There are constants $H < \infty$ and $\lambda_* > 0$, independent of k and of the target accuracy, such that every accepted iterate satisfies*

$$\text{res}_{F+B}(y_k) \leq \|d_k\| \leq \frac{H}{\Lambda_k} \leq \frac{H}{k\lambda_*}. \quad (44)$$

Let Q_N denote the total number of full trials used to produce the first N accepted iterates. Then

$$Q_N \leq N + m_\lambda + m_\eta(N) = N + O(\log(N+1)) + O(1) = O(N). \quad (45)$$

Consequently, for every $\varepsilon > 0$, the algorithm returns an ε -residual solution after at most

$$N_\varepsilon := \max \left\{ 1, \left\lceil \frac{H}{\lambda_*\varepsilon} \right\rceil \right\} = O(\varepsilon^{-1}) \quad (46)$$

accepted iterations and $O(\varepsilon^{-1})$ full trials.

In particular, the method uses $O(\varepsilon^{-1})$ evaluations of F and $O(\varepsilon^{-1})$ exact resolvent evaluations of B .

Proof. Lemmas 6 and 7 show that each line search terminates and that

$$\lambda_k \geq \lambda_* \quad \text{for every accepted } k.$$

Therefore,

$$\Lambda_k = \sum_{i=1}^k \lambda_i \geq k\lambda_*.$$

Combining this with Proposition 1 gives (44).

Every unsuccessful full trial reduces exactly one of the two retained parameters. There are at most m_λ primary reductions and at most $m_\eta(N)$ secondary reductions through accepted iteration N . Producing N accepted iterations therefore requires at most

$$N + m_\lambda + m_\eta(N)$$

full trials, which proves (45).

By (44),

$$\|d_{N_\varepsilon}\| \leq \varepsilon.$$

Thus the stopping test has triggered no later than accepted iteration N_ε . Since each full trial uses two evaluations of F and two exact resolvent evaluations of B , the claimed oracle bounds follow from (45). $\square$

8 Discussion of the oracle model

The preceding theorem is an oracle-operation result. It uses the convention, adopted in [3], that

$$J_{\gamma B} = (I + \gamma B)^{-1}$$

can be evaluated exactly at unit cost for every $\gamma > 0$.

The sufficient condition for the secondary search at iteration k is

$$\eta M \leq \frac{4\lambda\Delta}{(k+1)^2}.$$

Accordingly, the retained parameter η may reach a scale of order k^{-2} . The theorem does not assert that solving a proximal subproblem with such a parameter has bounded numerical bit complexity; it asserts that the number of calls to the stipulated exact-resolvent oracle is $O(\varepsilon^{-1})$.

The result also does not extend automatically to inexact resolvents. Such an extension would require controlling the resulting graph and Lyapunov errors and is a separate question from the removal of the logarithmic factor under the exact-resolvent model.

9 Conclusion

An anchored two-resolvent scheme with retained backtracking attains the last-iterate bound

$$\text{res}_{F+B}(y_k) = O(k^{-1})$$

when B is maximal monotone, $F + B$ is monotone with a nonempty zero set, and F is locally Lipschitz on $\text{cl}(\text{dom } B)$. Accounting for every rejected trial preserves the same order: the first N accepted iterates require only $N + O(\log N)$ trials. Hence an ε -residual solution is found in $O(\varepsilon^{-1})$ oracle operations.

This removes the logarithmic factor from the non-strongly monotone operation bound, answering the open question posed by Lu and Mei, under the same exact-resolvent oracle convention.

References

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